

Assignment 1: AI System Specification

Course: *Introduction to Artificial Intelligence*

Selected System: *AI Tutor for Personalized Student Learning Recommendations*

Task 1: PEAS Specification

System Name: AI Tutor for Personalized Student Learning Recommendations

Performance Measures

1. Student learning improvement rate.
2. Quiz and test accuracy score.
3. Course completion rate.
4. Student satisfaction level.
5. Recommendation relevance accuracy.

Environment

The AI Tutor operates in an online learning environment containing students, educational content, quizzes, assignments, and internet-based learning platforms. The environment continuously changes as students learn and interact with the system.

Actuators

Lesson recommendation engine, quiz generation system, feedback generator, notification service, and learning path adjustment module.

Sensors

Quiz scores, exam results, clickstream data, time spent on lessons, assignment submissions, and student feedback.

Task 2: Environment Classification

- 1. Partially Observable:** The AI Tutor cannot directly observe a student's actual understanding, concentration, or motivation.
- 2. Single-Agent:** The AI Tutor acts as the primary decision-making system that generates recommendations independently.
- 3. Stochastic:** Student behavior and learning outcomes are uncertain and may vary under similar conditions.
- 4. Sequential:** Current recommendations affect future learning progress and future recommendations.
- 5. Dynamic:** Student performance and learning needs continuously change over time.
- 6. Discrete:** The AI Tutor chooses from a finite set of lessons, quizzes, and learning resources.

Task 3: Critical Analysis

Question 1: *Which dimension poses the greatest technical challenge and why?*

The most challenging dimension is Partially Observable. An AI Tutor cannot directly measure a student's true understanding, attention level, confusion, or motivation. Instead, it must estimate these hidden factors using indirect indicators such as quiz scores, assignment results, and learning behavior. Because the available information is incomplete, the system may occasionally provide less effective recommendations, making this one of the most difficult challenges for AI engineers.

Question 2: Utility Function

Utility Function:

$$U = 0.7 \times \text{LearningAccuracy} - 0.3 \times \text{LearningTime}$$

This utility function encourages the AI Tutor to maximize learning accuracy while minimizing the amount of time required for learning. The goal is to provide efficient and effective recommendations.

If the weight of Learning Accuracy is doubled, the utility function becomes $U = 1.4 \times \text{LearningAccuracy} - 0.3 \times \text{LearningTime}$. In this case, the AI Tutor will prioritize accuracy more strongly and may recommend additional practice exercises, detailed explanations, and longer learning sessions to improve student understanding.

Structured Representation (JSON)

```
{
  "system_name": "AI Tutor for Personalized Student Learning Recommendations",
  "peas": {
    "performance_measures": [
      "student_learning_improvement_rate",
      "quiz_accuracy_score",
      "course_completion_rate",
      "student_satisfaction_level",
      "recommendation_relevance_accuracy"
    ],
    "environment": "Online learning platform containing students, educational content, quizzes, assignments",
    "actuators": [
      "lesson_recommendation_engine",
      "quiz_generation_system",
      "feedback_generator",
      "notification_service",
      "learning_path_adjustment_module"
    ],
    "sensors": [
      "quiz_scores",
      "exam_results",
      "clickstream_data",
      "time_spent_on_lessons",
      "assignment_submissions",
      "student_feedback"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "Student understanding and motivation cannot be directly observed."
    },
    "agents": {
      "choice": "Single-Agent",
      "justification": "The AI Tutor is the primary decision-making system."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Student responses and learning outcomes are uncertain."
    },
    "episodic_sequential": {
      "choice": "Sequential",
      "justification": "Current recommendations influence future learning outcomes."
    },
    "static_dynamic": {
      "choice": "Dynamic",
      "justification": "Student performance and learning needs change continuously."
    },
    "discrete_continuous": {
      "choice": "Discrete",
      "justification": "The system chooses from a finite set of learning resources and actions."
    }
  },
  "utility_function": "U = 0.7 * LearningAccuracy - 0.3 * LearningTime"
}
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